MAHADI HASSAN MUNNA

Full-Stack Software Engineer | AI & ML Enthusiast | Cyber Security

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LinkedIn | Portfolio | GitHub

PROFESSIONAL SUMMARY

Results-driven Software Engineer with 4 years of full-stack experience delivering robust, production-grade applications across web and mobile. Proven track record of steering complex initiatives from concept to stable release, achieving a 70% deployment rate with zero critical post-launch issues. Recognized by executive leadership with multiple awards for engineering excellence and innovation.

EDUCATION

Swiss Joint Master's in Computer Science (BENEFR) *Sep 2025 – Present*

University of Bern Bern, Switzerland

Bachelor's in Computer Science & Engineering *Feb 2017 – Apr 2021*

Military Institute of Science & Technology (MIST) Dhaka, Bangladesh

- GPA: 3.70/4.00 (Top 10% in department)
- **Thesis:** Development of a Hospital Cabin Management System Using IoT and Deep Learning Technologies (Supervisor: Dr. Mohammad Nazrul Islam)
- Relevant Coursework: Machine Learning, Mathematics, Algorithms, Operating Systems, Database Management

WORK EXPERIENCE

Software Engineer *Jul 2022 – Sep 2025*

IDLC Finance PLC Dhaka, Bangladesh

- Designed core enterprise software used daily by 4 internal subsidiaries, optimizing workflows and reducing task completion time by 40%.
- Developed a cross-platform mobile application integrating advanced AI features, amplifying user engagement by 25%.
- Modernized reporting infrastructure by migrating legacy tools to Power BI and SQL, achieving a 2× acceleration in report generation speed.
- Architected an AI agent toolkit using LMStudio, streamlining AI-driven workflows and boosting operational efficiency by 50%.
- Mentored Junior Software Engineers and Interns each year, providing technical guidance, accelerating onboarding, and improving overall team productivity.

Frontend Developer *Apr 2021 – Jun 2022*

Autoby24 GmbH Zurich, Switzerland

- **Frontend Leadership:** Pioneered the frontend development for two core platforms (Goby24 & Autoby24), delivering 90% of all feature implementations independently within one year.
- **Cross-Platform Development:** Rapidly prepared and deployed the Goby24 mobile ecosystem in under 3 months, achieving a 50% reduction in development cycles compared to standard estimates.

KEY PROJECTS

IDLC Customer App *Jan 2025 – Sep 2025*

React.js, Node.js, AI (MCP & RAG)

- Directed full-stack development, centralizing services across all subsidiaries and optimizing customer workflows, resulting in an 80% reduction in branch visits.
- Integrated advanced AI features using Model Context Protocol (MCP) and Retrieval-Augmented Generation (RAG) to improve customer experience.
- Redesigned database schema to accommodate escalating data volume and velocity from cross-platform mobile apps, decreasing query response time by 40%.

Mobile Recharge & Utility Bill Payment

Nov 2024 – Feb 2025

React.js, ASP.NET Core, API Integration

- Architected a full-scale web solution enabling seamless mobile recharges and utility bill payments, driving a 25% surge in user retention.
- Engineered comprehensive Client and Admin dashboards, ensuring secure transaction monitoring and a 100% mobile-responsive user experience.

Capital Market Back Office Solution

Aug 2022 – Dec 2024

React.js, ASP.NET Core, MS SQL Server, Power BI

- Implemented a robust back-office system facilitating critical financial transactions, lending services, and equity sales/trading, accelerating employees' daily task completion by 50%.
- Initiated robust data visualization and reporting infrastructure using Power BI Report Builder, doubling report generation speed.
- Optimized workflow processes and database queries, resulting in a 40% reduction in overall task completion time across internal support operations.

TECHNICAL SKILLS

Languages:	JavaScript, TypeScript, Python, C++, C#, C, Java, SQL
Web Development:	React.js, Next.js, Node.js, ASP.NET Core, Tailwind CSS, Bootstrap
Mobile Development:	React Native, Expo, Flutter, Android Studio
Database:	Microsoft SQL Server, PostgreSQL, Oracle Database
ML/AI:	PyTorch, TensorFlow, Scikit-learn, Keras, NumPy, Pandas, YOLO, RAG, MCP
Tools & Graphics:	Git, Docker, Power BI, Figma, Adobe XD, Blender, MATLAB, Google Colab
Testing:	Jest, Playwright

PUBLICATIONS

Islam, M.N., Aadeeb, M.S., Hassan Munna, M.M., et al.

A deep learning based multimodal interaction system for bedridden and immobile hospital admitted patients: design, development, and evaluation.

BMC Health Services Research (Q1), Springer Link, 2022

Islam, M.N., Aadeeb, M.S., Khan, R.R., Munna, M.M.H., Sarwar, M., Nasrin, S., Islam, A.N.

Developing a Novel Hands-Free Interaction Technique Based on Nose and Teeth Movements for Using Mobile Devices.

IEEE Access (Q1), Vol. 9, pp. 58127-58141, 2021

Aadeeb, M.S., Munna, M.M.H., Rahman, M.R., Islam, M.N.

Towards Developing a Hospital Cabin Management System using Brain-Computer Interaction.

20th International Conference on Intelligent Systems Design and Applications (ISDA 2020), 2020

KEY ACHIEVEMENTS

2023, 2024	**Chairman's Award:** Honored twice by C-level executives as a top-performing engineer among a 50+ member technical department, maintaining a 0% post-deployment rollback rate.
2022	**National Champion:** Selected from over 500 nationwide submissions as the winner of the prestigious BASIS National ICT Award for R&D Project of the Year.
2020	**Leadership:** Elected VP of Logistics for the MIST Computer Club, managing a 20-person team to organize 5 university-wide hackathons and tech symposiums serving over 1,000 students.
2018, 2019	**Algorithmic Competitor:** Participated consecutively two times in the National Collegiate Programming Contest (NCPC) representing MIST.

LANGUAGES

English:	Professional Working Proficiency (CEFR B2)
German:	Elementary Proficiency (CEFR A2)
Bengali:	Native / Bilingual Proficiency